

Session 1 Welcome to the IT World Explain careers in IT (developer, tester, designer, data analyst, AI engineer). Discuss product vs service companies; real-world examples (Google, Zoho, TCS). Overview of how teams collaborate using GitHub Slack, Jira. Trainer Demo: Show an open-source GitHub project (e.g., "Awesome-Java") and walk through contributors' roles. AI Tool Tip: Ask ChatGPT to list top emerging tech jobs (to discuss in class

Task –

- 1) Name 5 IT job roles and write one line about what each

Software Developer – Designs, writes, and maintains software applications and systems.

Software Tester (QA Engineer) – Tests software to find bugs and ensure it works correctly before release.

UI/UX Designer – Creates user-friendly and visually appealing interfaces for websites and applications.

1. **AI Engineer** – Develops artificial intelligence and machine learning models for smart applications and automation.

- 2) List 3 product companies and 3 service companies with one example of work each does.

3 Product Companies and 3 Service Companies

Company Type	Company	Example of Work
Product Company	Google	Develops products like Google Search, Gmail, and Android.
Product Company	Zoho	Creates business software such as Zoho CRM and Zoho Mail.
Product Company	Microsoft	Develops products like Windows, Office, and Azure.
Service Company	Tata Consultancy Services	Provides IT consulting, software development, and support services to clients.
Service Company	Infosys	Offers application development, cloud, and digital transformation services.
Service Company	Wipro	Provides IT services, cybersecurity, and business process outsourcing.

Difference:

- **Product Companies** build and sell their own software or products.
- **Service Companies** work on projects and solutions for other clients or businesses.

3) Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Full Forms and Uses

Term	Full Form	One-Line Use
GitHub	Git Hub (Git-based hosting platform)	Used to store, manage, and collaborate on source code projects.
Slack	Searchable Log of All Communication and Knowledge	Used for team communication, messaging, and collaboration.
Jira	(Originally derived from "Gojira"; not an official acronym)	Used to track tasks, bugs, and project progress in software development.
IDE	Integrated Development Environment	Software used by programmers to write, test, and debug code.
API	Application Programming Interface	Allows different software applications to

Term

Full Form

One-Line Use

communicate and share
data with each other.

Example:

- Developers store code on **GitHub**.
- Teams discuss work on **Slack**.
- Tasks are managed in **Jira**.
- Coding is done in an **IDE** like Visual Studio Code.
- Apps use **APIs** to exchange information.

- 4) Name any 5 programming languages and mention one popular use case for each.

**Programming
Language**

Popular Use Case

Python

Artificial Intelligence (AI) and Machine Learning applications.

Java

Android app development and enterprise software.

C++

Game development and high-performance applications.

JavaScript

Creating interactive websites and web applications.

C#

Developing Windows applications and games using Unity.

- 5) What is the difference between a developer and a tester? Answer in 2 lines

Developer: A developer writes and maintains software code to create applications and features.

Tester: A tester checks the software for bugs and defects to ensure it works correctly and meets requirements.

